



CIS 4951 Capstone II | Giggso

Team A: Critical Materials Recovery

Decision support for identifying valuable or critical materials in cataloged items.

Problem Understanding: Fragmented Data & Inefficient Material Assessment



OPERATIONAL CHALLENGE

Core Operational Gap

Organizations receiving surplus, obsolete, damaged, or unserviceable items lack an efficient method to determine if those items contain critical, strategic, precious, or valuable materials.

INFORMATION BOTTLENECK

Distributed Data Sources

Currently, relevant information is distributed across:

- Public item catalogs
- Technical descriptions
- Material references
- Lifecycle records
- Commodity publications

This fragmentation makes it difficult to quickly and accurately assess material composition.

Why This Matters: Operational & Compliance Risks of Manual Research



CORE CHALLENGE

Manual Research Bottlenecks

Manual research into material composition is slow and produces inconsistent results. This creates significant operational and compliance risks.

Relying on ungrounded manual processes or unsupported AI inferences without a traceable source introduces severe operational exposure across the organization.

Missed Recovery & Retention Opportunities

Missed opportunities to retain useful items, recover scarce materials, or recycle valuable content.

Inconsistent Analytical Results

Slow, manual evaluations yield variable and non-standardized material assessments.

Poor Recommendation Documentation

Manual processes consistently fail to properly document why a specific recommendation was made.

Lack of Verifiable Proof & Auditability

Inability to provide traceable audit trails or objective proof for compliance checks.

Risk of Unsupported AI Assumptions

Relying on ungrounded AI inferences without a traceable source introduces severe operational risk.

EXPECTED OUTCOME

Traceable Decision-Support System Prototype

Building a reproducible prototype to automate item matching, material analysis, and disposition recommendations with full provenance.

CORE SYSTEM MANDATE

Every system recommendation must explicitly expose underlying evidence, decision rules, uncertainty metrics, and source provenance.

PROTOTYPE ARCHITECTURE & PIPELINE FLOW



01 / INGESTION

Catalog Matching

Accepts a public item identifier or description and automatically retrieves matching public catalog information.

02 / ANALYSIS

Critical Mineral Mapping

Connects item data to material information and compares components with current critical-material references.

03 / DECISION ENGINE

Action Recommendations

Recommends specific lifecycle paths:

- Retention or Reclamation
- Recycling or Manual Review
- Ordinary Disposal

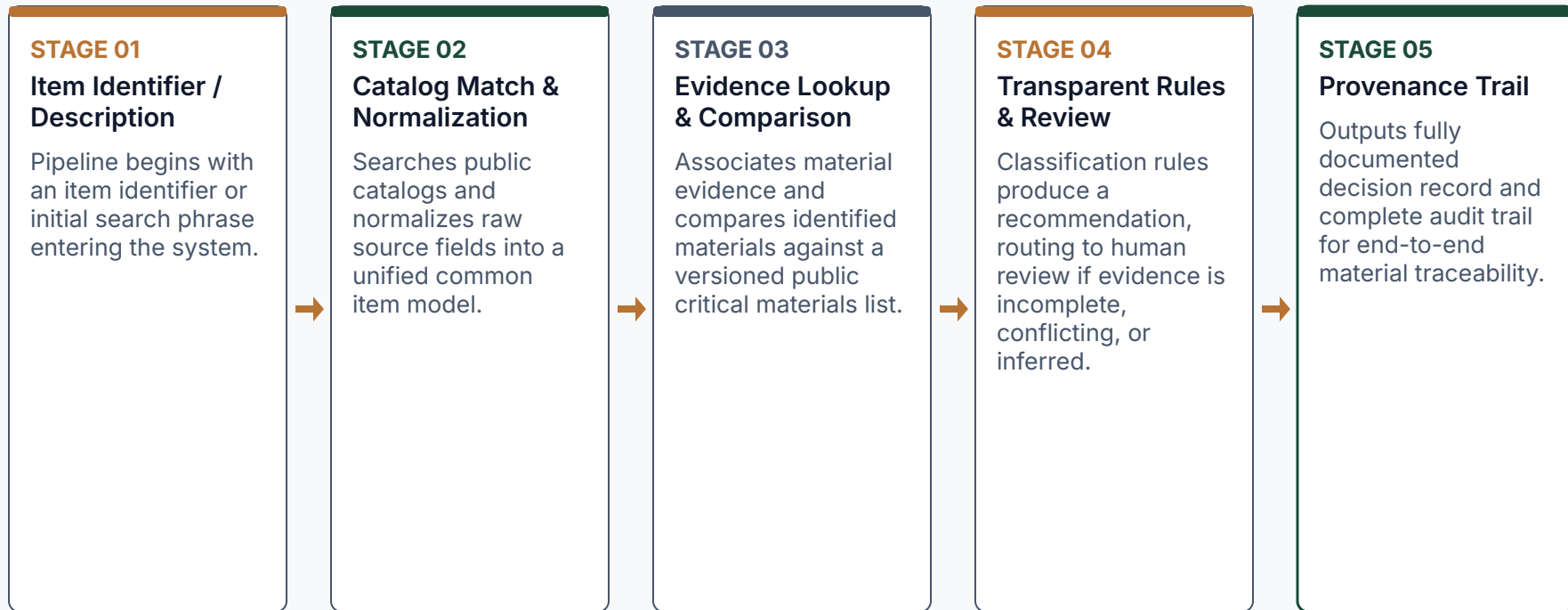
04 / TRACEABILITY

Evidence & Provenance

Every recommendation explicitly presents supporting evidence, underlying rules, uncertainty metrics, and source provenance.

Core Workflow

End-to-End Decision & Provenance Pipeline



Key Data Entities: Core Structural & Provenance Graph Model



Category 01

Core Objects

- **Item:** Physical asset tracked across lifecycle.
- **Component:** Sub-assembly or structural element.
- **Material:** Base substance and material composition.

Category 02

Evidence & References

- **Composition Evidence:** Verified empirical analytical records.
- **Criticality Reference:** Standardized risk and importance classification.

Category 03

Financial & Action

- **Lifecycle Event:** Historical operational transitions.
- **Recovery Method:** Defined material extraction pathway.
- **Value Estimate:** Illustrative economic valuation models.

Category 04

Audit & Governance

- **Recommendation:** Automated decision guidance engine.
- **Review Decision:** Human-in-the-loop authorization entry.
- **Provenance Record & Audit Event:** Full trace log history.

Data Sources: Public, Reference & Synthetic Datasets



PUBLIC LOGISTICS

PUB LOG

Serves as a key source for non-restricted item identifiers, baseline metadata, and logistics characteristics within the prototype.

MINERAL REFERENCES

USGS Geological Datasets

- **USGS Critical Minerals List:** Versioned criticality references.
- **USGS Mineral Commodity Summaries:** Broader supply & commodity context.

MATERIALS & SYNTHETIC DATA

Repositories & Demo Records

- **NIST Materials Data Repository:** Standardized materials data integration.
- **EPA RCRAInfo:** Optional environmental compliance tracking source.
- **Synthetic Datasets:** Generated records for testing & validation.

IMPORTANT LICENSING CAVEAT

PUB LOG Licensing Notice

PUB LOG is treated as **publicly releasable data** within this prototype, but it is **not automatically described as openly licensed** unless explicit licensing terms permit.

Proposed Technical Approach

Data Normalization & Rule-Based Inference



STAGE 01 — DATA MODELING

Catalog Data Normalization

Initial technical build begins by normalizing raw source fields from disparate supplier and internal catalogs into a unified, common item model.

STAGE 02 — MATERIAL VERIFICATION

Critical Minerals Comparison

All identified materials within the normalized model are systematically cross-referenced and evaluated against a versioned critical-minerals repository to maintain compliance and supply security.

Transparent Recommendation Rules

The system applies clear, deterministic rule logic to process item data and generate automated material handling recommendations.

Mandatory Human Review Triggers

EXPLICIT OVERRIDE CONDITIONS:

- Material composition is inferred rather than explicit
- Evidence or underlying catalog data is incomplete
- Source datasets or catalog records conflict
- A potentially hazardous material is identified

Immutable Audit Trail

Complete, permanent logging of all raw inputs, source records, and active rules applied for full system traceability.

Architecture & Tooling: Roles of Aryx and Raven



DATA & CONTEXT LAYER

Aryx

Serves as the candidate context and data layer supporting key architectural representations:

- Knowledge graph representation
- Provenance tracking
- Entity relationships required for item-material mapping

&

DEV WORKFLOW GUARDRAIL

Raven

Acts as a development workflow guardrail maintaining engineering discipline:

- Reducing AI comprehension debt
- Executing security checks
- Maintaining documentation discipline

SCOPE BOUNDARY & SYSTEM INTEGRITY

Neither tool replaces the project-specific recommendation logic required for material classification.

Human Review & Explainability

Transparent Evidence Display & Decision Framework



01 | ITEM IDENTIFICATION

Matched Item Record

Canonical database entry linked to current specimen analysis.

SKU: MAT-2026-8941

Catalog Ref: Structural Alloy B-2

02 | COMPOSITION SPECTRUM

Identified Materials

Detected physical components & element composition.

Primary: Recycled Aluminum (68%)

Secondary: Copper Substrate (24%)

Trace: Polymeric Coating (8%)

03 | SYSTEM ACCURACY

Confidence Level

94.8%

HIGH SYSTEM CONFIDENCE

04 | PROVENANCE & LOGIC TRACE

Supporting Evidence & Classification Rules

Applied Rule: Rule #14-B (Spectral Signature & Density Threshold Matching)

Evidence Source: Multi-point NIR spectrometry & XRF surface scan.

ATTENTION: CONFLICTING DATA DETECTED

Surface coating thickness (8%) conflicts with standard Grade A specs.
Explicit verification flagged.

05 | DECISION CONTROL & AUDIT

Human Action & Review Triggers

HUMAN CONFIRMATION REQUIRED

Reviewers must select an action and provide a mandatory justification reason:

ACCEPT
Confirm Match
REVISE
Edit Class/Data
REJECT
Flag Incorrect

Evaluation Plan: Quantifiable Metrics & Benchmark Dataset



100+

Public or Synthetic Item Records

Benchmark Dataset Target

Precision / Recall

Material Identification Rate

Core Accuracy Metric

100%

Complete Provenance Goal

Recommendation Auditability

Dataset & Automated Identification

- **Benchmark Records:** Minimum of 100 public or synthetic records featuring fully documented expected outcomes.
- **Catalog-Match Success:** Quantitative rate measuring exact matches against cataloged items.
- **Material-ID Performance:** Strict evaluation using precision and recall metrics for raw material categorization.

Human Review & Operational Efficiency

- **Uncertain Case Routing:** Percentage of low-confidence or ambiguous cases correctly routed to human reviewers.
- **Reviewer Agreement & Speed:** Tracking inter-reviewer agreement rate alongside overall average processing times.
- **Data Limitation Documentation:** Explicitly logging known edge cases, missing data, and system constraints.



Sprint 1 Plan: Delivering an End-to-End Thin Slice

STEP 01

Alignment & Item Selection

- **PO Alignment:** Confirm the smallest acceptable demo with the Product Owner.
- **Sample Selection:** Select 5–10 public or synthetic item examples for initial validation.
- **Scope Boundaries:** Establish baseline criteria for the sprint demo artifact.



STEP 02

Schema & Source Setup

- **Initial Schema:** Define entities for Items, Materials, Evidence, Criticality References, Recommendations, and Provenance.
- **Source Integration:** Configure one primary public source alongside a synthetic fallback.



STEP 03

Execution & Evaluation

- **Thin-Slice Build:** Deliver one end-to-end evidence-backed recommendation sample.
- **Substrate Eval:** Evaluate Aryx as our core graph substrate.
- **Backlog Readiness:** Complete Odoo ticket preparation for ongoing workstreams.



Initial Questions for Product Owner (Giggso)

CATEGORY 01

Data Priorities

Catalog Source Hierarchy

Confirm the explicit priority order for integration across catalog sources:

1. PUB LOG
2. USGS
3. NIST

CATEGORY 02

System Scope

Sprint 1 Demo Size

Define the smallest acceptable baseline scope for the initial Sprint 1 demonstration.

Recovery Value Estimates

Clarify whether illustrative recovery value estimates should be included in the early project phase.

CATEGORY 03

Governance

Evidence Quality

Establish a standardized evidence-quality labeling scheme across data entities.

Human Reviewer Persona

Define the intended persona and workflow expectations for the human reviewer.

Repository Boundaries

Establish public repository data boundaries and security guardrails.

Team Contribution & Workstream Allocation



Scoping & Research

- Requirements scoping
- Data source research



Schema & Provenance Architecture

- Schema design
- Provenance design



Implementation & Evaluation

- Prototype implementation
- Developing evaluation plan
- Creating presentations
- Tracking Odoo tickets

GROUP COORDINATION & COMMUNICATION

Group coordination is maintained by sharing daily project hours and working schedules to ensure consistent progress and communication.

FINAL SUMMARY

Conclusion

Traceable, Evidence-Backed
Decision Support

PROJECT TARGET

Zero Unsupported AI
Assumptions

"Team A's mission is to deliver a traceable, reproducible decision-support prototype connecting item records to verifiable material evidence."

- MISSION STATEMENT

The system will not treat an AI-generated inference as a confirmed material composition unless it is supported by a traceable source.

By relying on transparent rules and explicit human review thresholds, we will mitigate the risks of unsupported AI guesses and manual research inefficiencies.

